DETECTION OF QUASI-COHERENT MODULES BY PULLBACK ALONG AN ETALE COVER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the trivial first implication, the descent of the hypothesis to a scheme chart by composing etale surjections, the production of global presentations on every object of the site from local presentations of the pullback, and the reduction of the last condition to affine charts.

Editorial note. The label lemma-characterize-quasi-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-properties.tex –label lemma-characterize-quasi-coherent –start 5608 –end 5655 –local-id 28052 returned RESULT=OK (L28052 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Quasi-coherence of the pullback of a quasi-coherent module. Same chapter, the lemma that the pullback of a quasi-coherent module along an etale morphism of algebraic spaces is quasi-coherent. Supplies the passage from condition (2) to condition (3).
- Local presentations of quasi-coherent modules on a site. Modules on Sites, the definition of a globally presented module, the lemma producing local presentations, and the criterion for a module on a site to be quasi-coherent. Supplies the construction of global presentations on the members of an etale covering and the resulting quasi-coherence.

Source reference. Original equation 03LW: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 4629–4629. Prerequisite pointer invoked at proof line(s) 5636; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01BG: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 5461–5461. Prerequisite pointer invoked at proof

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

line(s) 5631; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 03DE: sources/stacks-chow/tags.txt, source lines 4417–4417. Prerequisite pointer invoked at proof line(s) 5640; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Prerequisite pointer invoked at proof line(s) 5649; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03DN: sources/stacks-chow/tags.txt, source lines 4426–4426. Prerequisite pointer invoked at proof line(s) 5641; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$-modules. The following are equivalent*

1. *$\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module,*
2. *there exists an étale morphism $f : Y \to X$ of algebraic spaces over S with $|f| : |Y| \to |X|$ surjective such that $f^*\mathcal{F}$ is quasi-coherent on Y,*
3. *there exists a scheme U and a surjective étale morphism $\varphi : U \to X$ such that $\varphi^*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_U$-module, and*
4. *for every affine scheme U and étale morphism $\varphi : U \to X$ the restriction $\varphi^*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_U$-module.*

Proof. It is clear that (1) implies (2) by considering id_X. Assume $f : Y \to X$ is as in (2), and let $V \to Y$ be a surjective étale morphism from a scheme towards Y. Then the composition $V \to X$ is surjective étale as well and by Lemma 01BG the pullback of $\mathcal{F}$ to V is quasi-coherent as well. Hence we see that (2) implies (3).

Let $U \to X$ be as in (3). Let us use the abuse of notation introduced in Equation (03LW). As $\mathcal{F}|_{U_{étale}}$ is quasi-coherent there exists an étale covering $\{U_i \to U\}$ such that $\mathcal{F}|_{U_{i,étale}}$ has a global presentation, see Modules on Sites, Definition 03DE and Lemma 03DN. Let $V \to X$ be an object of $X_{étale}$. Since $U \to X$ is surjective and étale, the family of maps $\{U_i \times_X V \to V\}$ is an étale covering of V. Via the morphisms $U_i \times_X V \to U_i$ we can restrict the global presentations of $\mathcal{F}|_{U_{i,étale}}$ to get a global presentation of $\mathcal{F}|_{(U_i \times_X V)_{étale}}$ Hence the sheaf $\mathcal{F}$ on $X_{étale}$ satisfies the condition of Modules on Sites, Definition 03DL and hence is quasi-coherent.

The equivalence of (3) and (4) comes from the fact that any scheme has an affine open covering. □